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Q1.1 – Why do thread IDs appear in different orders?

The threads execute concurrently, so the operating system scheduler determines when each thread receives CPU time. Because scheduling depends on CPU availability, thread state, and system activity, the order in which threads reach the printf statement can change between executions.

Q1.2 – What happens when P exceeds the number of CPU cores?

When the number of software threads exceeds the available physical CPU cores, the CPU must switch between threads. These context switches introduce overhead because the operating system must save and restore thread state. Too many threads can therefore reduce performance instead of improving it. Cache effects can also become less efficient.

Q1.3 – What is the barrier?

At the end of an OpenMP parallel region, there is an implicit barrier. All worker threads must finish the parallel region before execution continues. This prevents the master thread from proceeding to subsequent code while other threads are still performing the previous parallel work.

Q1.4 – Hardware thread vs OS thread vs virtual thread

Type	Meaning
Hardware thread	Execution context provided by the CPU, e.g. SMT/Hyper-Threading
OS thread	Thread managed and scheduled by the operating system
Virtual/green thread	Lightweight software-managed thread that can be multiplexed onto OS threads

Task 1.2 – Thread Oversubscription Results

Number of Threads (P)	Execution Time (seconds)
1	0.000035048
2	0.000070095
4	0.000117779

8	0.000166893
16	0.000581026
32	0.000891924
64	0.001546144

Task 1.3 results

Thread s	Execution time (seconds)
1	0.053941
2	0.054044
4	0.043785
8	0.071783